Guaranteed versus Experienced False-Alert Rates: Evaluating Conformal Selection for Provincial Dengue Outbreak Bulletins on Held-out Countries

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Abstract. Dengue early-warning systems issue monthly bulletins listing the provinces expected to exceed an epidemic threshold, and are trusted only if few alerts are false. Conformal selection, which applies the Benjamini–Hochberg (BH) procedure to conformal p-values, can promise in advance that the expected share of false alerts *per bulletin* stays below a level q. A health officer, however, experiences the share of false alerts among *all* alerts received, which is a different quantity. We measure this difference with hypotheses fixed before the confirmation data were analysed: monthly counts for 337 provinces in nine countries, probabilistic forecasts 1–3 months ahead, five development countries and four held-out confirmation countries (the Philippines, Viet Nam, Sri Lanka and Brazil; 178 provinces). At $q = 0.2$ the guarantee holds (bulletin false discovery rate 0.12–0.14, with or without the origins that lacked calibration data), but the false-alert share users receive is 0.225, 0.277 and 0.332 at 1, 2 and 3 months ahead, above q at 2 and 3 months. The gap between the two rates is 0.09–0.17 larger than that of a fixed-threshold rule, whose own two gap components largely offset each other. On the experienced scale the fixed-threshold rule has a lower false-alert share without lower recall; on the guaranteed scale conformal selection has higher recall at 2–3 months (exploratory). The gap is tied to how alerts group into months: spreading the same alerts over months at random removes most of it. Agencies should therefore decide which false-alert rate a bulletin promises before choosing the alerting rule.

Keywords: Conformal Selection · False Discovery Rate · Early Warning System · Dengue · Outbreak Detection · Pre-registration

1 Introduction

1.1 Motivation

Dengue is the fastest-spreading mosquito-borne viral disease, and South and South-East Asia carry most of its burden. Provincial health departments decide

one to three months ahead where to spray, stock fluids and open surge beds, helped by early-warning bulletins that flag provinces expected to exceed an endemic threshold [8,14,19]. A bulletin is trusted only if it is rarely wrong, because every false alarm costs vector-control resources and credibility.

1.2 Problem Statement and Research Gap

Conformal selection [16] offers a finite-sample promise of the kind an agency would like to announce: "on average, at most 20% of the flagged provinces will be false alarms". The false discovery rate (FDR) in that sentence is $\mathbb{E}[V/\max(R,1)]$ over bulletins, where V is the number of false alerts and R the number of alerts in one bulletin; a bulletin without alerts contributes zero. A health officer instead experiences the share of false alerts among all alerts received over a season, $\sum V/\sum R$, the marginal FDR [12, 21]. Theory says the two can differ; what is missing is (i) how large the difference is on real multi-country surveillance data, (ii) whether it changes which alerting rule an agency should prefer, and (iii) where it comes from.

1.3 Research Questions and Hypotheses

This study answers three research questions (RQ) and tests five pre-registered hypotheses (H1–H5, numbered as in the pre-registration; H2b refines H2), all at $q = 0.2$:

- **RQ1.** Does conformal selection keep its per-bulletin guarantee on held-out countries, and how far is the experienced (pooled) FDR from it? **(H1)** The upper 95% CI of the bulletin FDR is at most q at every horizon. **(H2)** The gap $\Delta = \mathrm{FDR}_{\mathrm{pool}} - \mathrm{FDR}_{\mathrm{b}}$ has a lower CI above 0 at every horizon. **(H2b)** The gap of conformal selection exceeds that of a pre-specified fixed-threshold rule.
- **RQ2.** Which rule is preferable on the experienced scale? **(H4)** Conformal selection is dominated on (pooled FDR, recall) by a member of a fixed-threshold family at two or more horizons.
- **RQ3.** Can the gap be reduced or labelled? **(H3)** An epidemic-state Mondrian calibration lowers the pooled FDR. **(H5)** A trailing-precision label meets its 90% bound in at least 85% of country-years.

1.4 Contributions

The contributions of this work are:

C1 A pre-registered[3] evaluation of conformal selection for multi-province outbreak bulletins on nine countries, with a development/confirmation split by country and a fixed mapping from outcomes to claims.

C2 A quantified gap between the guaranteed and the experienced FDR, with paired country-year bootstrap confidence intervals and an exact decomposition into an empty-bulletin part and an alert-weighting part.

[3] Code, data sources, the frozen pre-registration and all result files: https://github.com/tranquanguit/dengue-calibrated-alerts.

C3 An analysis of where the gap comes from: two permutation references tie it to the month-level grouping of alerts, and a simulation illustrates how forecast errors shared within a month produce such grouping.

C4 Transparent negative results (Mondrian calibration, trailing-precision labels and, on development data only, Learn-then-Test certification) and a reporting protocol for health agencies.

2 Related Work

2.1 Background

The BH procedure [6] controls $\mathbb{E}[V/\max(R,1)]$ under independence and positive dependence [7]. When R is random, this differs from the positive FDR $\mathbb{E}[V/R \mid R > 0]$ [20] and from the marginal FDR $\mathbb{E}[V]/\mathbb{E}[R]$ [12], for which the optimal rule is a single global threshold on a local-fdr statistic [21]. Conformal p-values [5, 22] need no distributional assumption; combined with BH they select units whose outcome exceeds a threshold with finite-sample FDR control [16]. Extensions address non-exchangeable data [4,13], general risks [1,2] and sequences of selections [15, 24].

2.2 Prior Work and Limitations

Dengue forecasting studies evaluate point or probabilistic accuracy [9, 11, 17]. Early-warning systems such as EWARS [14] and the Brazilian prototype [19] turn forecasts into alerts with an endemic-channel threshold [8] and report detection skill, but announce no false-alert promise. Conformal selection supplies such a promise, yet its evaluations report only the guaranteed quantity. To our knowledge, no study has measured the guaranteed and the experienced FDR side by side on real outbreak bulletins, or asked whether the choice between them changes the recommended rule.

3 Methodology

3.1 Overview

The pipeline has four stages (Fig. 1): monthly provincial counts, probabilistic forecasts taken from a companion study (seven quantiles, 1–3 months ahead), alerting rules that turn the forecasts into a monthly bulletin, and an evaluation of every rule on both FDR notions.

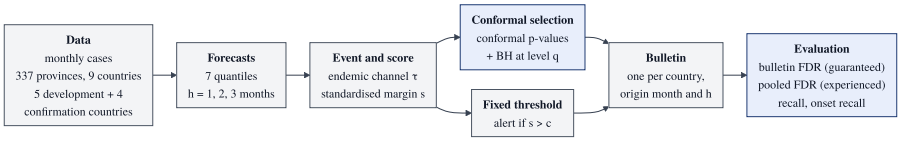


Fig. 1. Pipeline of the study. Each alerting rule (conformal selection, the fixed-threshold family, and the CRC and top-k baselines) issues one bulletin per country, origin month and horizon; every bulletin is scored on both FDR notions.

3.2 Data and Data Pipeline

Monthly dengue counts at the first sub-national level come from OpenDengue V1.3 [10] and national portals. The Viet Nam series (60 provinces, 1998–2010) was compiled by the authors from public sources, with split or merged provinces re-aggregated into continuous series. Missing months are left missing (2.9% of province-months in 2006–2010, 0.3% in 2015–2024). Countries were split before any analysis into a development set, used to design the procedures, and a confirmation set, run once after the hypotheses were frozen (Table 1). The data are aggregated public counts without identifiable information.

Table 1. Countries, roles and event rates (units = provinces or states)

Development				Confirmation			
Country	Units	Test years	Events	Country	Units	Test years	Events
Thailand	77	2015–2022	16.0%	Philippines	65	2006–2010	35.1%
Indonesia	32	2015–2022	18.0%	Viet Nam	60	2006–2010	17.4%
Cambodia	24	2006–2010	16.3%	Sri Lanka	26	2016–2024	23.6%
Laos	16	2006–2010	20.5%	Brazil	27	2017–2024	17.4%
Malaysia	10	2006–2010	24.8%				

Event. A province-month is an *endemic-channel exceedance* when $\log(1 + \text{cases})$ exceeds the mean plus two standard deviations of the same calendar month in the previous five years (highest year excluded), with at least five cases [8]. An *onset* is an event month whose previous month is not an event.

Forecasts. On the confirmation set the forecasts are the companion study's ensemble: a per-quantile average, on the log scale, of a LightGBM model pooled across countries [18] and the zero-shot foundation model Chronos-2 [3]. LightGBM hyper-parameters are tuned only on years before each test year, and Chronos-2 is not trained. The risk score is a standardised margin,

$$s = \frac{\log(1 + \hat{q}_{0.5}) - \tau}{\max\{\log(1 + \hat{q}_{0.9}) - \log(1 + \hat{q}_{0.1}),\, 0.05\}}, \tag{1}$$

where τ is the log endemic threshold; a cell is *eligible* if $\hat{q}_{0.5} \geq 5$ cases.

3.3 Alerting Procedures

A *bulletin* is one (country, origin month, horizon h).

Conformal selection (CS). At origin t the calibration set holds the eligible cells of the same country and horizon whose target month is already observed, with at least 200 cells. For an eligible cell j with score s_j,

$$p_j = \frac{1 + \#\{i \leq n :\ Y_i = 0,\ s_i \geq s_j\}}{n + 1}, \tag{2}$$

where $Y_i = 0$ marks a past non-event; ineligible cells get $p_j = 1$. Alerts are the BH selection at level q. If calibration and test cells are exchangeable given eligibility, and the bulletin's cells are independent given the calibration set, then

$\mathbb{E}[V/\max(R,1)] \le q$ [16]. Shared forecast errors violate exactly this exchangeability, so we treat the assumption as the object of study.

Remedy. Mondrian CS restricts calibration to past cells with the same national epidemic state (at least 20% of provinces above their channel at the origin). It was the single remedy pre-registered for confirmation. Learn-then-Test [1] and adaptive level feedback [13] were tried on development data only.

Baselines. A frozen fixed-threshold family alerts when $s > c$ (or a margin without standardisation exceeds c), with $c \in \{-0.3, \ldots, 0.5\}$; the pre-specified comparator is the margin rule with $c = 0$, i.e. "alert when the median forecast exceeds the threshold". A conformal-risk-control-style scan (CRC) [2] and a top-k rule are descriptive baselines.

3.4 Evaluation Protocol

For every rule we report three FDR notions over the B bulletins: the guaranteed bulletin FDR, the FDR of bulletins that contain alerts, and the experienced (pooled) FDR,

$$\mathrm{FDR}_{\mathrm{b}} = \tfrac{1}{B}\textstyle\sum_b V_b/\max(R_b,1), \tag{3}$$

$$\mathrm{FDR}_{>0} = \mathrm{mean}\{V_b/R_b : R_b > 0\}, \tag{4}$$

$$\mathrm{FDR}_{\mathrm{pool}} = \textstyle\sum_b V_b \,/\, \sum_b R_b, \tag{5}$$

together with recall of event months, onset recall and lead time. The gap $\Delta = \mathrm{FDR}_{\mathrm{pool}} - \mathrm{FDR}_{\mathrm{b}}$ decomposes exactly as

$$\Delta = \underbrace{(\mathrm{FDR}_{>0} - \mathrm{FDR}_{\mathrm{b}})}_{\text{empty bulletins}} + \underbrace{\frac{\mathrm{Cov}(R, \mathrm{FDP} \mid R > 0)}{\mathbb{E}[R \mid R > 0]}}_{\text{alert weighting}}, \tag{6}$$

where the second term is positive when large bulletins are more often wrong. All confidence intervals (95%) come from the pre-registered country-year block bootstrap (1,000 resamples), with paired resamples for comparisons between rules; numbers are computed from unrounded values. A country-stratified bootstrap and leave-one-country-out estimates serve as robustness checks.

4 Experimental Setup

The hypotheses H1–H5 and an outcome-to-claim table were frozen in three versions on timestamped GitHub branches before the confirmation set was analysed. Analyses outside the plan are labelled exploratory: the burn-in-excluded results, both permutation references, the leave-one-country-out and stratified estimates, the comparison at a matched bulletin FDR and the Learn-then-Test runs. The main deviation is that origins without enough calibration cells (32–40 of 317 bulletins per horizon) were scored as empty bulletins for CS, whereas the plan excluded them; since this lowers the bulletin FDR and recall of CS only, main results are also given without these burn-in origins. The 317 bulletins per horizon are 27 country-years of 12 months, less seven months after May 2024 without Brazilian data. A simulation (40 provinces, 14 years, 300 replicates) adds a forecast error w shared by all cells of a bulletin, with the share of error variance set

to that of the confirmation forecasts (0.23–0.28 on average; 0.05–0.08 for Viet Nam, 0.42–0.54 for Sri Lanka).

5 Results

5.1 Guarantee and Experienced FDR (RQ1)

Table 2 summarises the confirmation results at $q = 0.2$. The guarantee holds: the bulletin FDR of CS is 0.119–0.126 with upper CIs of 0.158–0.161 (H1 supported), and 0.137–0.140 with upper CIs of 0.175–0.179 when the burn-in origins are excluded. The experienced FDR is 0.225, 0.277 and 0.332 at $h = 1, 2, 3$. Against the announced level it is indistinguishable from q at $h = 1$ (CI 0.176–0.286) and exceeds it at $h = 2$ and $h = 3$. Against the bulletin FDR that CS actually achieves, the gap Δ is 0.100, 0.153 and 0.212, with lower CIs of 0.062, 0.094 and 0.140 (H2 supported); 68–76% of it is the empty-bulletin part.

Table 2. Confirmation results at $q = 0.2$, pooled over the four held-out countries. $\mathrm{FDR_b}$: bulletin (guaranteed) FDR; $\mathrm{FDR_{pool}}$: pooled (experienced) FDR; $\Delta = \mathrm{FDR_{pool}} - \mathrm{FDR_b}$ = empty-bulletin part + alert-weighting part, Eq. (6). 95% CIs from the country-year block bootstrap; burn-in excl.: origins without enough calibration cells removed (exploratory).

Rule	h	$\mathrm{FDR_b}$ [CI]	$\mathrm{FDR_{pool}}$ [CI]	Δ [CI]	Empty	Alert-wt.	Recall
Conformal selection (CS)	1	0.126 [0.095, 0.158]	0.225 [0.176, 0.286]	0.100 [0.062, 0.155]	0.068	0.032	0.644
	2	0.124 [0.090, 0.160]	0.277 [0.211, 0.360]	0.153 [0.094, 0.239]	0.117	0.036	0.454
	3	0.119 [0.082, 0.161]	0.332 [0.246, 0.433]	0.212 [0.140, 0.316]	0.155	0.057	0.326
CS, burn-in excl.	1	0.140 [0.108, 0.175]	0.225 [0.176, 0.286]	0.086 [0.052, 0.138]	0.054	0.032	0.687
	2	0.140 [0.105, 0.177]	0.277 [0.211, 0.360]	0.137 [0.080, 0.219]	0.101	0.036	0.488
	3	0.137 [0.099, 0.179]	0.332 [0.246, 0.433]	0.195 [0.122, 0.298]	0.138	0.057	0.354
Mondrian CS	1	0.134 [0.103, 0.164]	0.217 [0.170, 0.280]	0.083 [0.046, 0.137]	0.069	0.014	0.616
	2	0.128 [0.092, 0.165]	0.271 [0.200, 0.355]	0.142 [0.082, 0.225]	0.112	0.030	0.412
	3	0.126 [0.089, 0.166]	0.330 [0.245, 0.426]	0.204 [0.130, 0.307]	0.146	0.058	0.312
Fixed threshold, $c = 0$	1	0.162 [0.134, 0.189]	0.168 [0.135, 0.212]	0.006 [−0.022, 0.043]	0.031	−0.025	0.682
	2	0.214 [0.177, 0.249]	0.236 [0.186, 0.301]	0.022 [−0.015, 0.077]	0.054	−0.032	0.526
	3	0.236 [0.193, 0.281]	0.283 [0.226, 0.358]	0.047 [0.003, 0.108]	0.069	−0.022	0.395

The fixed-threshold rule $c = 0$ has a much smaller gap, but not because the mechanism is absent: its empty-bulletin part (0.031–0.069) is largely offset by a negative alert-weighting part, and at $h = 3$ its gap is 0.047 [0.003, 0.108]. The paired difference $\Delta_{\mathrm{CS}} - \Delta_{\mathrm{threshold}}$ is 0.093 [0.066, 0.129], 0.131 [0.096, 0.170] and 0.165 [0.120, 0.220] (H2b supported). It stays positive without the burn-in origins (0.076, 0.116, 0.146; lower CIs 0.054, 0.081, 0.102) and when any one country is left out (range 0.086–0.186). The alert-weighting part is positive for CS and negative for the threshold rule: when CS issues many alerts in one month, those alerts are more often wrong.

5.2 Which Rule Is Preferable (RQ2)

Fig. 2 traces each rule family across its tuning range. On the experienced scale (top row) the fixed-threshold family lies above CS at every horizon, and the pre-registered dominance test is met at all three horizons (H4 supported). Against

the pre-specified comparator $c = 0$, the pooled FDR of CS is higher by 0.057 [0.022, 0.093], 0.041 [0.004, 0.074] and 0.048 [−0.003, 0.094], and its recall differs by −0.038 [−0.110, 0.014], −0.072 [−0.154, −0.007] and −0.069 [−0.140, −0.010]. Without the burn-in origins the pooled-FDR excess is 0.067, 0.053 and 0.060 (all CIs exclude 0) and the recall differences (−0.003 to −0.054) are not significant. At a matched pooled FDR the threshold rule also detects more epidemic onsets (0.465, 0.401, 0.346 against 0.340, 0.234, 0.189). On the guaranteed scale (bottom row) the ranking reverses: at a matched bulletin FDR, CS has higher recall at $h = 2$ (+0.115 [0.054, 0.159]) and $h = 3$ (+0.125 [0.067, 0.173]), +0.058 and +0.090 without the burn-in origins, and ties at $h = 1$ (exploratory). This is what theory predicts: the bulletin FDR rewards alert-free months, which BH produces in quiet periods, whereas a single global threshold is the optimal form for the marginal FDR [21].

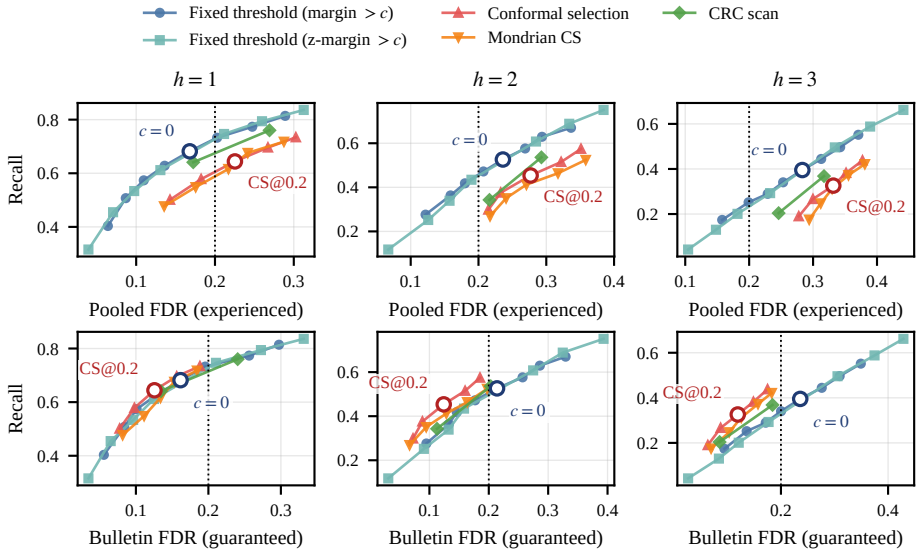


Fig. 2. Precision–recall frontiers on the four confirmation countries. Top: experienced (pooled) FDR, where the fixed-threshold family lies above conformal selection. Bottom: guaranteed (bulletin) FDR, where conformal selection lies on or above the threshold frontier. Threshold curves trace c from −0.3 to 0.5; CS and Mondrian trace q from 0.1 to 0.3. The dotted line marks $q = 0.2$.

5.3 Where the Gap Comes From

Two permutation references break the month-level grouping (exploratory, burn-in-excluded origins). First, keeping the alerts CS actually issued and only shuffling them across the bulletins of the same country and year (sizes kept, BH not re-run) leaves the pooled FDR unchanged and shrinks the gap from 0.086, 0.137 and 0.195 to −0.016, 0.016 and 0.050 (s.d. over 20 shuffles ≤ 0.008), while the bulletin FDR of the regrouped alerts rises to 0.24–0.28, above q. CS thus

meets its per-bulletin guarantee because its alerts concentrate in some months while other bulletins stay empty. Second, re-running BH on the same p-values regrouped into random pseudo-bulletins gives a gap of 0.037, 0.084 and 0.102, so 57%, 39% and 48% of the gap depends on the real grouping even when the alert set may change. Month-level grouping reflects both forecast errors shared within a month and epidemic seasonality, which these references cannot separate. The simulation (Fig. 3) illustrates the first mechanism: under i.i.d. errors CS is exact on its own quantity (0.204) yet has a pooled FDR of 0.306, and the gap grows with the shared error. When the forecast misses a year-level shock, the bulletin FDR of CS itself exceeds q (0.227). The simulated threshold rule works at a different false-alert level, so the simulation says nothing about the ranking of rules.

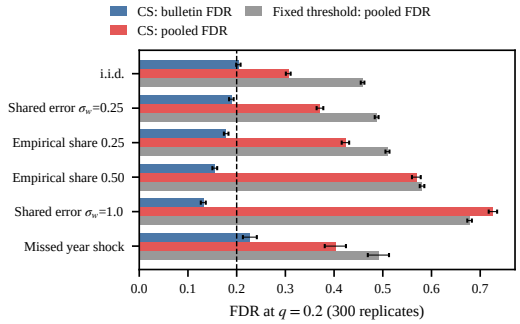


Fig. 3. Simulation at $q = 0.2$ (bars: ± 1.96 Monte Carlo s.e.). The guaranteed FDR of CS stays at or below q except when the forecast misses a year-level shock, while its experienced FDR grows with the error shared within a bulletin.

5.4 Remedies and Labels (RQ3)

The pre-registered Mondrian test concerns the pooled FDR, and Mondrian calibration does not lower it: the paired difference from CS is -0.008 $[-0.025, 0.010]$, -0.006 $[-0.037, 0.022]$ and -0.002 $[-0.021, 0.020]$ (H3 not supported). It narrows the gap slightly at $h = 1$ (-0.016 $[-0.024, -0.008]$; CIs include 0 at $h = 2, 3$), but by raising its bulletin FDR rather than lowering the experienced one, and at a cost in recall (-0.028 $[-0.051, -0.007]$ at $h = 1$). On the development set only, Learn-then-Test with Hoeffding or empirical-Bernstein bounds issued at most 303 alerts out of several thousand eligible cells, too few to be useful, and adaptive level feedback traded recall for precision. A trailing 24-month precision label met its 90% lower bound in only 68–69% of country-years (H5 not supported), because precision drifts between epidemic years.

6 Discussion

H1 and H2 are supported: conformal selection keeps its per-bulletin promise on held-out countries, also without the burn-in origins, yet the false-alert share that users receive is 0.10–0.21 above the bulletin FDR it achieves and above q at

$h = 2, 3$. **H2b is supported:** the gap is much larger under per-bulletin BH than under a fixed threshold, whose empty-bulletin and alert-weighting parts largely cancel. **H4 is supported:** on the experienced scale a simple threshold rule is the better operating choice, while on the guaranteed scale conformal selection is. **H3 and H5 are not supported:** the Mondrian state does not lower the experienced FDR, and a naive precision label is over-confident.

Practical implications. The choice of FDR notion decides the method. We recommend that agencies (1) state whether the promise is per bulletin or per alert received; (2) publish, next to any conformal level, the realised pooled false-alert share and its year-to-year spread; (3) not present a trailing-precision interval as a guarantee; and (4) compare any calibrated rule with a fixed threshold on the frontier of the chosen notion.

Limitations. The confirmation set has about 27 country-years, with only five test years for the Philippines and Viet Nam, so the intervals are wide; the stratified bootstrap and leave-one-country-out analysis lead to the same conclusions. The plan was frozen on Git branches rather than in a registry. The forecasts come from a single pipeline, the event is a statistical channel rather than a clinical threshold, reporting delays are not modelled, and the permutation references cannot separate shared forecast errors from seasonality. Online and batch FDR procedures [15,24] and betting-based bounds [23] were not evaluated.

7 Conclusion and Future Work

On four held-out countries and under a pre-registered plan, conformal selection kept its per-bulletin false-alert guarantee, but the false-alert share experienced by bulletin users was 0.09–0.17 further from it than under a fixed-threshold rule, which was preferable on the experienced scale. The gap is tied to how alerts concentrate in some months, which both shared forecast errors and epidemic seasonality produce. Alert systems should state which false-alert rate they control and be evaluated on that rate's frontier. Future work includes online FDR procedures over the sequence of bulletins, external validation on other regions, and a monthly-updated warning dashboard.

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